# Build Your Own **Miniature VAW Meter**

Fayaz Hassan

EFY TESTED S.C. Dwivedi

The circuit presented here can measure voltage (V), current (A) and wattage (W) on an OLED. It is actually voltmeter, ammeter and wattmeter circuits combined in one unit. It is a useful tool for testing circuits, desktop power supplies, battery chargers, mini inverters and so on.

Problems associated with a multimeter include changing positions of probes while measuring voltage and current, inability to measure voltage and current simultaneously, chances of multimeter getting damaged if voltage is measured in high current mode and others. This circuit solves all these problems by displaying all measured values simultaneously on the OLED.

## Circuit and working

The circuit diagram of the miniature VAW meter is shown in Fig. 1. It is built around 5V voltage regulator 7805 (IC1), microcontroller ATtiny85 (IC2), 30A current sensing module ACS712, miniature display (2.4cm, or 0.96-inch, in size) OLED and a few other components.

The circuit can measure maximum voltage up to 30V DC and maximum current up to 30A DC. To avoid damaging the microcontroller (MCU), measurements should be limited to 30V and 30A. If the value exceedes 30V, the display shows OFLO, which indicates overflow of the parameter.

***ATtiny85.*** ATtiny85 is an eight-pin MCU sufficiently powerful for the project, with 8kB flash, 512B RAM, 512B EEPROM and 1MHz default clock frequency at the heart of the system. It continuously measures voltage and current through its two ADC channels. ADC values are processed and converted to voltage, current and power values, and then displayed on the OLED.

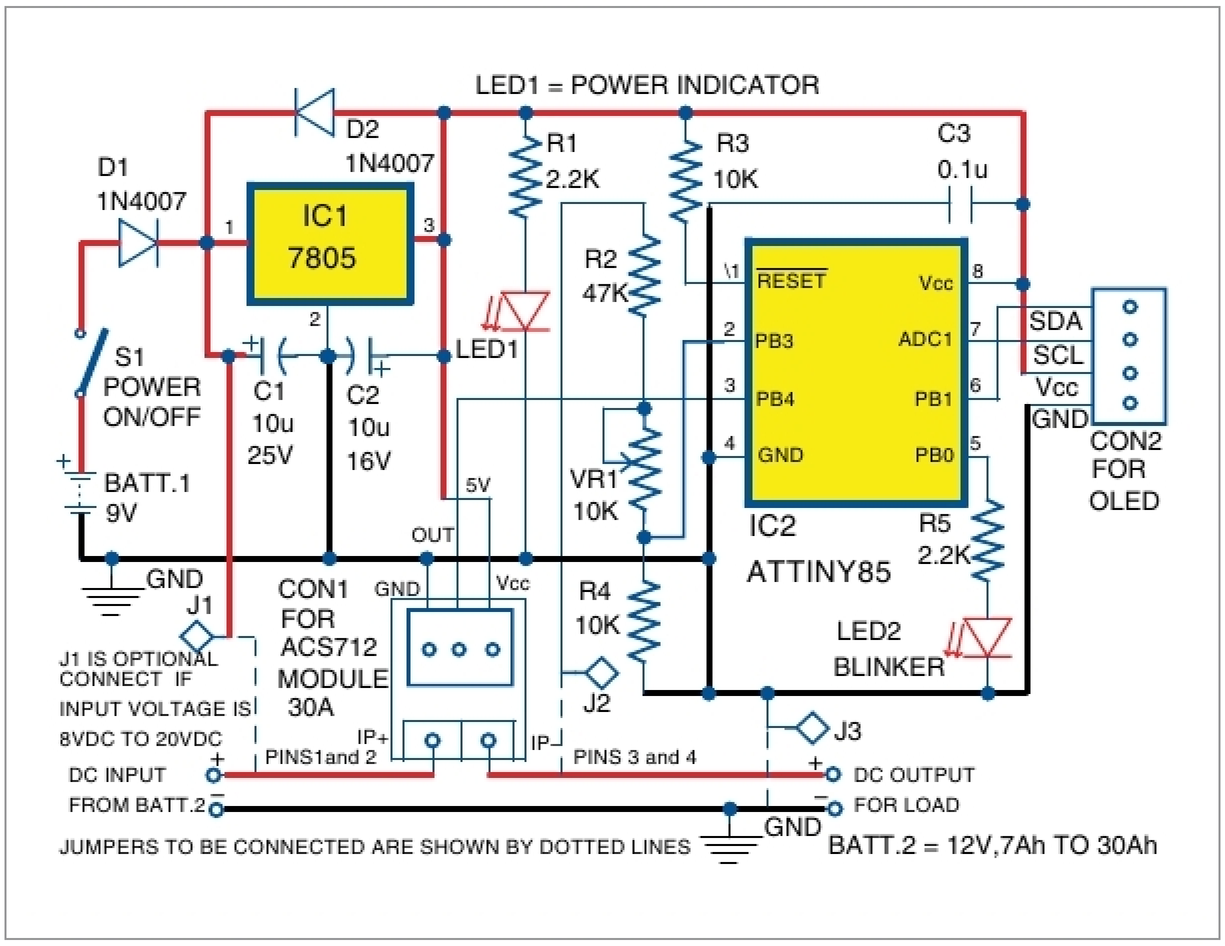


*Fig. 1: Circuit diagram of the miniature VAW meter*

***ACS712.*** The current-sensing module uses Allegro's ACS712 chip, which uses Hall effect to convert current to voltage. Output of the module is fed to one of the ADC channels of the MCU. A 30A module is used for measuring current. Sensitivity of the module is 66mV/A (refer ACS712 data sheet).


*Fig. 2: OLED display*

***OLED.*** It is a small display measuring 2.4 cms with 128 × 64-pixel resolution, 5V DC supply and I2C connector. It has four pins, out of which one is for serial data (SDA pin), another for clock (SCL pin) and the remaining two for power supply.

To show text on the display, 64 rows of pixels are divided into eight text rows with eight-pixel rows in each text row. To display text clearly, code is set for sixteen-pixel height and eight-pixel width. So, OLED can display text in four rows with 16 × 8 font size.

**EFY Note**
The source code of this project is available for free download at *source.efymag.com*